Edge-Linking-using-Hough-Transformm

Aim:

To write a Python program to detect the lines using Hough Transform.

Software Required:

Anaconda - Python 3.7

Algorithm:

Step1:

Import all the necessary modules for the program.

Step2:

Load a image using imread() from cv2 module.

Step3:

Convert the image to grayscale.

Step4:

Using Canny operator from cv2,detect the edges of the image.

Step5:

Using the HoughLinesP(),detect line co-ordinates for every points in the images.Using For loop,draw the lines on the found co-ordinates.Display the image.

Program

Input image and grayscale image

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread('Tiger.jpg')  # Replace with your image path
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Input Image')
plt.axis('off')
```

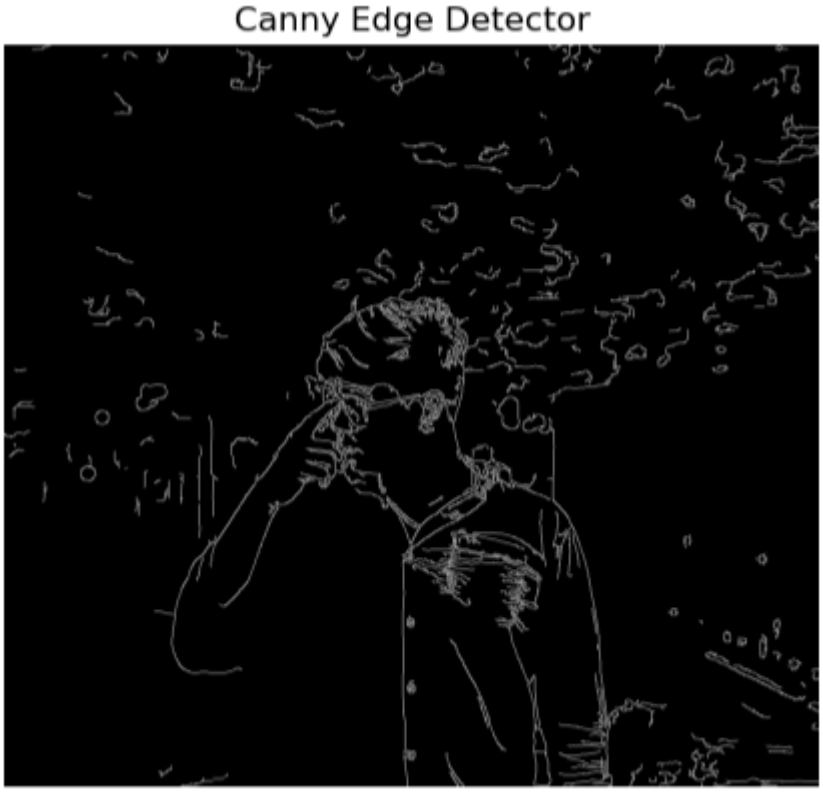
Input Image

```
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
plt.imshow(gray_image, cmap='gray')
plt.title('Grayscale Image')
plt.axis('off')
```

Grayscale Image

Canny Edge detector output

```
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
plt.imshow(gray_image, cmap='gray')
plt.title('Grayscale Image')
plt.axis('off')
```



Display the result of Hough transform

```
lines = cv2.HoughLinesP(edges, rho=1, theta=np.pi/180, threshold=100, minLineLength=50,
maxLineGap=10)
output_image = image.copy()

if lines is not None:
    for line in lines:
        x1, y1, x2, y2 = line[0]
        cv2.line(output_image, (x1, y1), (x2, y2), (0, 255, 0), 2)
plt.imshow(cv2.cvtColor(output_image, cv2.COLOR_BGR2RGB))
plt.title('Hough Transform - Line Detection')
plt.axis('off')
```

Result of Hough Transform

Result:

Thus,The Python program to detect the lines using Hough Transform run successfully.